```
$ ls -la
$ docker ps -a
$ kubectl get pods
$ git status
$ terraform plan
$ _
```

aws

docker

git

kubernetes

Jenkins

PLAN
↓
CODE
↓
BUILD
↓
TEST
↓
DEPLOY
↓
MONITOR

TOP 500

• MOST USED •

DEVOPS COMMANDS

```
$_
```

DEV OPS

• BUILD • AUTOMATE • DEPLOY • MONITOR

YOUR ULTIMATE DEVOPS COMMAND REFERENCE

CI/CD PIPELINE

AUTOMATE → INTEGRATE → DELIVER → OPERATE

```
$ git clone <repo>
$ docker build -t app .
$ kubectl get pods
$ aws s3 ls
```

DevOps Shack

Top 500 Most Used DevOps Commands

A Complete Command Reference for Linux, Git, Docker, Kubernetes, Helm, Terraform, Ansible, AWS, Azure, CI/CD, Networking, Monitoring & Security

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Table of Contents

Introduction

This reference compiles 500 of the most frequently used commands across the modern DevOps toolchain — from core Linux administration to container orchestration, infrastructure automation, and cloud platform management. Each command is presented with a concise, practical explanation of what it does and when to use it.

Whether you are preparing for a DevOps interview, building daily muscle memory at the terminal, or simply need a fast lookup while troubleshooting production systems, this document is designed to be a single, organized source of truth.

Commands are grouped into 16 categories covering the full DevOps lifecycle: source control, containers, orchestration, infrastructure as code, configuration management, public cloud (AWS & Azure), CI/CD, networking, observability, and security.

1. Linux Fundamentals & File System

Core Linux commands every DevOps engineer uses daily for navigating, inspecting, and managing the file system.

Command	Description
`ls -la`	List all files (including hidden) in long format with permissions, owner, size, and date.
`pwd`	Print the current working directory.
`cd /path/to/dir`	Change the current directory.
`mkdir -p dir1/dir2`	Create nested directories in one step, no error if they exist.
`rmdir dirname`	Remove an empty directory.
`rm -rf dirname`	Forcefully and recursively delete a directory and its contents.
`cp -r src dest`	Copy a directory recursively.
`mv old new`	Move or rename a file or directory.
`touch file.txt`	Create an empty file or update its timestamp.
`cat file.txt`	Print file contents to standard output.
`tac file.txt`	Print file contents in reverse line order.
`less file.txt`	View file contents page by page.
`head -n 20 file.txt`	Show the first 20 lines of a file.
`tail -n 20 file.txt`	Show the last 20 lines of a file.
`tail -f /var/log/syslog`	Follow a log file in real time as it's written.
`find / -name '*.log'`	Search the filesystem for files matching a pattern.
`find . -mtime -1`	Find files modified in the last 24 hours.
`locate filename`	Quickly locate files using a prebuilt index.
`grep -rn 'ERROR' /var/log/`	Recursively search for a pattern with line numbers.
`grep -i 'error' file.log`	Case-insensitive search for a pattern.
`awk '{print $1}' file.txt`	Print the first column/field of each line.
`sed 's/foo/bar/g' file.txt`	Replace all occurrences of 'foo' with 'bar'.
`sort file.txt`	Sort lines of a file alphabetically.
`sort -n file.txt`	Sort lines numerically.
`uniq file.txt`	Remove adjacent duplicate lines.
`wc -l file.txt`	Count the number of lines in a file.
`diff file1 file2`	Show line-by-line differences between two files.
`chmod 755 file.sh`	Change file permissions (owner rwx, group/others rx).

Command	Description
`chmod +x script.sh`	Make a script executable.
`chown user:group file`	Change file owner and group.
`ln -s /target /linkname`	Create a symbolic link.
`tar -czvf archive.tar.gz dir/`	Create a compressed tar archive.
`tar -xzvf archive.tar.gz`	Extract a compressed tar archive.
`zip -r archive.zip dir/`	Create a zip archive of a directory.
`unzip archive.zip`	Extract a zip archive.
`du -sh dir/`	Show total disk usage of a directory in human-readable form.
`df -h`	Show disk space usage of mounted filesystems.
`stat file.txt`	Display detailed file metadata (inode, permissions, timestamps).
`file file.txt`	Determine the type of a file.
`readlink -f file`	Resolve the absolute path of a symbolic link.
`xargs`	Build and execute commands from standard input.
`watch -n 2 'df -h'`	Run a command repeatedly every 2 seconds.

2. Process, System & Resource Monitoring

Commands to inspect running processes, system resources, and service health on Linux servers.

Command	Description
`ps aux`	List all running processes with detailed info.
`ps -ef | grep nginx`	Filter running processes by name.
`top`	Interactive real-time view of system processes and resource usage.
`htop`	Enhanced interactive process viewer with color UI.
`kill -9 PID`	Forcefully terminate a process by its PID.
`killall processname`	Kill all processes matching a name.
`pkill -f pattern`	Kill processes matching a command-line pattern.
`nice -n 10 command`	Run a command with adjusted CPU scheduling priority.
`renice -n 5 -p PID`	Change the priority of a running process.
`free -h`	Display memory usage in human-readable format.
`vmstat 2`	Report virtual memory statistics every 2 seconds.
`iostat -x 2`	Report CPU and I/O device statistics.
`uptime`	Show system uptime and load averages.
`uname -a`	Display kernel and system information.
`lscpu`	Display detailed CPU architecture information.
`lsblk`	List block devices (disks and partitions).
`lsof -i :8080`	List processes using a specific network port.
`netstat -tulnp`	Show listening ports and associated processes.
`ss -tulnp`	Modern replacement for netstat to show sockets.
`systemctl status nginx`	Check the status of a systemd service.
`systemctl start nginx`	Start a systemd-managed service.
`systemctl stop nginx`	Stop a systemd-managed service.
`systemctl restart nginx`	Restart a systemd-managed service.
`systemctl enable nginx`	Enable a service to start on boot.
`systemctl daemon-reload`	Reload systemd unit files after changes.
`journalctl -u nginx -f`	Follow live logs for a specific systemd unit.
`journalctl --since today`	Show system logs since the start of today.
`dmesg | tail`	Show recent kernel ring buffer messages.
`crontab -e`	Edit the current user's scheduled cron jobs.

Command	Description
`crontab -l`	List the current user's cron jobs.
`whoami`	Print the current logged-in user.
`id`	Show user and group ID information.
`history | grep ssh`	Search command history for a pattern.
`env`	Display all current environment variables.
`export VAR=value`	Set an environment variable for the session.
`alias ll='ls -la'`	Create a shortcut for a longer command.
`nohup command &`	Run a command immune to hangups, in the background.
`screen -S session`	Start a persistent terminal session.
`tmux new -s session`	Start a new tmux multiplexed terminal session.
`strace -p PID`	Trace system calls made by a running process.
`uptime -p`	Show uptime in a pretty human-readable form.

3. Git Version Control

Essential Git commands for source control, branching, and collaboration in CI/CD pipelines.

Command	Description
`git init`	Initialize a new Git repository.
`git clone <url>`	Clone a remote repository to local machine.
`git status`	Show the working directory and staging area state.
`git add .`	Stage all changed files for commit.
`git add -p`	Interactively stage specific changes (hunks).
`git commit -m 'message'`	Commit staged changes with a message.
`git commit --amend`	Modify the most recent commit.
`git push origin main`	Push committed changes to a remote branch.
`git push --force`	Force push, overwriting remote history (use with caution).
`git pull origin main`	Fetch and merge changes from the remote branch.
`git fetch --all`	Fetch updates from all remotes without merging.
`git branch`	List all local branches.
`git branch -a`	List all local and remote branches.
`git checkout -b feature/x`	Create and switch to a new branch.
`git switch feature/x`	Switch to an existing branch (modern syntax).
`git merge feature/x`	Merge a branch into the current branch.
`git rebase main`	Reapply commits on top of another base branch.
`git rebase -i HEAD~3`	Interactively rebase the last 3 commits.
`git cherry-pick <hash>`	Apply a specific commit onto the current branch.
`git stash`	Temporarily shelve uncommitted changes.
`git stash pop`	Reapply the most recently stashed changes.
`git stash list`	List all stashed change sets.
`git log --oneline --graph`	Show commit history as a compact graph.
`git log -p -2`	Show the last 2 commits with full diffs.
`git diff`	Show unstaged changes in the working directory.
`git diff --staged`	Show staged changes ready for commit.
`git reset --soft HEAD~1`	Undo the last commit, keeping changes staged.
`git reset --hard HEAD~1`	Undo the last commit and discard all changes.
`git revert <hash>`	Create a new commit that undoes a previous commit.

Command	Description
`git tag v1.0.0`	Create a tag for a release point.
`git push --tags`	Push all local tags to the remote.
`git remote -v`	List configured remotes with their URLs.
`git remote add origin <url>`	Add a new remote repository.
`git clean -fd`	Remove untracked files and directories.
`git blame file.txt`	Show who last modified each line of a file.
`git show <hash>`	Show metadata and diff for a specific commit.
`git submodule update --init`	Initialize and update Git submodules.
`git config --global user.name 'name'`	Set the global Git username.
`git bisect start`	Binary search commits to find a regression.
`git worktree add ../path branch`	Check out a branch into a separate working directory.

4. Docker — Container Management

Commands for building, running, and managing Docker containers and images.

Command	Description
`docker --version`	Show the installed Docker version.
`docker run -d -p 80:80 nginx`	Run a container in detached mode with port mapping.
`docker run -it ubuntu bash`	Run a container interactively with a terminal.
`docker ps`	List currently running containers.
`docker ps -a`	List all containers, including stopped ones.
`docker stop <container>`	Gracefully stop a running container.
`docker start <container>`	Start a previously stopped container.
`docker restart <container>`	Restart a container.
`docker rm <container>`	Remove a stopped container.
`docker rm -f <container>`	Force remove a running container.
`docker images`	List all locally stored images.
`docker rmi <image>`	Remove a Docker image.
`docker build -t myapp:1.0 .`	Build an image from a Dockerfile in the current directory.
`docker pull nginx:latest`	Pull an image from a registry.
`docker push myrepo/myapp:1.0`	Push an image to a remote registry.
`docker tag myapp:1.0 myrepo/myapp:1.0`	Tag an image for a registry.
`docker logs -f <container>`	Stream logs from a container.
`docker exec -it <container> bash`	Open an interactive shell inside a running container.
`docker inspect <container>`	Show detailed JSON metadata about a container.
`docker stats`	Show live resource usage stats for running containers.
`docker network ls`	List Docker networks.
`docker network create mynet`	Create a custom Docker network.
`docker network inspect mynet`	Inspect details of a Docker network.
`docker volume ls`	List Docker volumes.
`docker volume create myvol`	Create a named Docker volume.
`docker volume inspect myvol`	Show details about a Docker volume.
`docker-compose up -d`	Start services defined in docker-compose.yml in the background.
`docker-compose down`	Stop and remove docker-compose services and networks.

Command	Description
`docker-compose logs -f`	Follow logs from all compose services.
`docker-compose build`	Build images defined in a compose file.
`docker-compose ps`	List containers managed by docker-compose.
`docker system prune -a`	Remove all unused containers, images, and networks.
`docker cp file.txt <container>:/path`	Copy a file from host into a container.
`docker save -o image.tar myimage`	Export a Docker image to a tar archive.
`docker load -i image.tar`	Import a Docker image from a tar archive.
`docker history myimage`	Show the build history/layers of an image.
`docker diff <container>`	Show filesystem changes made inside a container.
`docker top <container>`	Show running processes inside a container.
`docker commit <container> newimage`	Create a new image from a container's current state.
`docker login`	Authenticate to a Docker registry.
`docker context ls`	List available Docker contexts (remote hosts).
`docker buildx build --platform linux/amd64,linux/arm64 .`	Build multi-architecture images.
`docker swarm init`	Initialize a Docker Swarm cluster.
`docker service create --name web nginx`	Create a service in a Docker Swarm cluster.
`docker secret create mysecret file.txt`	Create a Swarm secret from a file.
`docker events`	Stream real-time events from the Docker daemon.
`docker port <container>`	Show port mappings for a container.
`docker pause <container>`	Pause all processes within a container.
`docker unpause <container>`	Resume a paused container.
`docker update --memory 512m <container>`	Update resource limits of a running container.
`docker info`	Display system-wide Docker information.

5. Kubernetes — Cluster & Workload Management

Comprehensive kubectl commands for managing Kubernetes clusters, workloads, and resources — the backbone of modern container orchestration.

Command	Description
`kubectl version`	Show the client and server Kubernetes versions.
`kubectl cluster-info`	Display cluster endpoint and core component info.
`kubectl get nodes`	List all nodes in the cluster.
`kubectl get nodes -o wide`	List nodes with extended details (IP, OS, kernel).
`kubectl describe node <node>`	Show detailed information about a specific node.
`kubectl get pods`	List pods in the current namespace.
`kubectl get pods -A`	List pods across all namespaces.
`kubectl get pods -o wide`	List pods with extended info (node, IP).
`kubectl describe pod <pod>`	Show detailed info and events for a pod.
`kubectl logs <pod>`	Show logs from a pod's container.
`kubectl logs -f <pod>`	Stream live logs from a pod.
`kubectl logs <pod> -c <container>`	Show logs from a specific container in a multi-container pod.
`kubectl exec -it <pod> -- bash`	Open an interactive shell inside a pod.
`kubectl delete pod <pod>`	Delete a specific pod.
`kubectl apply -f deployment.yaml`	Create or update resources defined in a manifest.
`kubectl create -f deployment.yaml`	Create resources from a manifest (fails if it exists).
`kubectl delete -f deployment.yaml`	Delete resources defined in a manifest.
`kubectl get deployments`	List deployments in the current namespace.
`kubectl describe deployment <name>`	Show detailed info about a deployment.
`kubectl rollout status deployment/<name>`	Watch the rollout status of a deployment.
`kubectl rollout history deployment/<name>`	Show revision history of a deployment.
`kubectl rollout undo deployment/<name>`	Roll back a deployment to its previous revision.
`kubectl rollout restart deployment/<name>`	Restart all pods in a deployment.

Command	Description
kubectl scale deployment <name> --replicas=5	Scale a deployment to a specific replica count.
kubectl autoscale deployment <name> --min=2 --max=10 --cpu-percent=80	Configure horizontal pod autoscaling.
kubectl get rs	List ReplicaSets.
kubectl get svc	List services in the current namespace.
kubectl describe svc <svc>	Show detailed info about a service.
kubectl expose deployment <name> --port=80 --type=LoadBalancer	Create a service to expose a deployment.
kubectl get ingress	List ingress resources.
kubectl describe ingress <name>	Show detailed info about an ingress resource.
kubectl get configmap	List ConfigMaps.
kubectl create configmap myconfig --from-file=config.txt	Create a ConfigMap from a file.
kubectl get secrets	List Secrets.
kubectl create secret generic mysecret --from-literal=key=value	Create a Secret from literal values.
kubectl get namespaces	List all namespaces in the cluster.
kubectl create namespace dev	Create a new namespace.
kubectl config get-contexts	List all available kubeconfig contexts.
kubectl config use-context <context>	Switch the active kubeconfig context.
kubectl config set-context --current --namespace=dev	Set the default namespace for the current context.
kubectl get events --sort-by='.lastTimestamp'	List cluster events sorted by time.
kubectl top pods	Show CPU/memory usage for pods (requires metrics-server).
kubectl top nodes	Show CPU/memory usage for nodes.
kubectl get pv	List PersistentVolumes.
kubectl get pvc	List PersistentVolumeClaims.
kubectl describe pvc <name>	Show detailed info about a PersistentVolumeClaim.
kubectl get statefulsets	List StatefulSets.
kubectl get daemonsets	List DaemonSets.

Command	Description
kubectl get jobs	List Kubernetes Jobs.
kubectl get cronjobs	List Kubernetes CronJobs.
kubectl port-forward pod/<pod> 8080:80	Forward a local port to a pod's port.
kubectl cp <pod>:/path ./local	Copy files from a pod to local machine.
kubectl label pod <pod> env=prod	Add a label to a resource.
kubectl annotate pod <pod> note='info'	Add an annotation to a resource.
kubectl taint nodes <node> key=value:NoSchedule	Apply a taint to a node to repel pods.
kubectl cordon <node>	Mark a node as unschedulable.
kubectl uncordon <node>	Mark a node as schedulable again.
kubectl drain <node> --ignore-daemonsets	Safely evict all pods from a node for maintenance.
kubectl edit deployment <name>	Open a resource definition in an editor for live editing.
kubectl patch deployment <name> -p '{"spec":{"replicas":3}}'	Apply a strategic merge patch to a resource.
kubectl get crd	List CustomResourceDefinitions.
kubectl explain pod.spec	Show documentation for a resource's fields.
kubectl api-resources	List all available resource types in the cluster.
kubectl get all -n <namespace>	List all resources in a given namespace.
kubectl set image deployment/<name> <container>=<image>	Update the container image of a deployment.
kubectl auth can-i create pods	Check if current user can perform an action.
kubectl get roles -n <namespace>	List RBAC Roles in a namespace.
kubectl get rolebindings -n <namespace>	List RBAC RoleBindings in a namespace.
kubectl get clusterroles	List cluster-wide RBAC roles.
kubectl get networkpolicies	List NetworkPolicy resources.
kubectl debug node/<node> -it --image=busybox	Launch a debug pod on a node for troubleshooting.

Command	Description
`kubectl get hpa`	List HorizontalPodAutoscaler resources.
`kubectl wait --for=condition=Ready pod/<pod>`	Wait for a resource to reach a specific condition.

6. Helm — Kubernetes Package Management

Helm commands for packaging, deploying, and managing Kubernetes applications via charts.

Command	Description
`helm version`	Show the installed Helm version.
`helm repo add bitnami https://charts.bitnami.com/bitnami`	Add a chart repository.
`helm repo update`	Refresh local cache of chart repositories.
`helm search repo nginx`	Search repositories for a chart.
`helm install myapp bitnami/nginx`	Install a chart as a new release.
`helm install myapp ./mychart -f values.yaml`	Install a local chart with custom values.
`helm upgrade myapp ./mychart`	Upgrade an existing release with a new chart version.
`helm upgrade --install myapp ./mychart`	Install the chart if not present, otherwise upgrade.
`helm rollback myapp 1`	Roll back a release to a specific revision.
`helm uninstall myapp`	Uninstall a release.
`helm list`	List all releases in the current namespace.
`helm list -A`	List releases across all namespaces.
`helm status myapp`	Show the status of a deployed release.
`helm get values myapp`	Show the values used for a release.
`helm get manifest myapp`	Show the rendered Kubernetes manifests for a release.
`helm history myapp`	Show revision history of a release.
`helm template ./mychart`	Render chart templates locally without installing.
`helm lint ./mychart`	Validate a chart for syntax and best-practice issues.
`helm create mychart`	Scaffold a new chart directory structure.
`helm package ./mychart`	Package a chart directory into a .tgz archive.
`helm dependency update ./mychart`	Download chart dependencies defined in Chart.yaml.

7. Terraform — Infrastructure as Code

Terraform commands for provisioning and managing cloud infrastructure declaratively.

Command	Description
`terraform init`	Initialize a working directory and download providers.
`terraform plan`	Preview changes Terraform will make to infrastructure.
`terraform apply`	Apply changes to reach the desired infrastructure state.
`terraform apply -auto-approve`	Apply changes without an interactive approval prompt.
`terraform destroy`	Destroy all resources managed by the configuration.
`terraform validate`	Check configuration files for syntax correctness.
`terraform fmt`	Reformat configuration files to canonical style.
`terraform show`	Display the current state in human-readable form.
`terraform state list`	List resources tracked in the current state.
`terraform state show <resource>`	Show attributes of a specific resource in state.
`terraform state rm <resource>`	Remove a resource from state without destroying it.
`terraform state mv <src> <dst>`	Move/rename a resource within state.
`terraform import <resource> <id>`	Import an existing resource into Terraform state.
`terraform output`	Show output values from the root module.
`terraform workspace list`	List available Terraform workspaces.
`terraform workspace new dev`	Create a new workspace.
`terraform workspace select dev`	Switch to a different workspace.
`terraform taint <resource>`	Mark a resource for recreation on the next apply.
`terraform refresh`	Reconcile state with real-world infrastructure.
`terraform graph`	Generate a visual dependency graph of resources.
`terraform console`	Open an interactive console to evaluate expressions.
`terraform providers`	List providers required by the configuration.
`terraform get`	Download and update modules referenced in configuration.
`terraform plan -out=plan.tfplan`	Save a plan to a file for later apply.
`terraform apply plan.tfplan`	Apply a previously saved plan file.

Command	Description
`terraform force-unlock <lock-id>`	Manually unlock a stuck state lock.
`terraform plan -destroy`	Preview a destroy operation without executing it.
`terraform login`	Authenticate with Terraform Cloud.
`terraform version`	Show the installed Terraform version.
`terraform plan -var='env=prod'`	Pass a variable value directly on the command line.

8. Ansible — Configuration Management & Automation

Ansible commands for agentless configuration management, application deployment, and orchestration.

Command	Description
`ansible --version`	Show the installed Ansible version.
`ansible all -m ping`	Test connectivity to all hosts in inventory.
`ansible-playbook site.yml`	Run a playbook against the inventory.
`ansible-playbook site.yml --check`	Perform a dry-run without making changes.
`ansible-playbook site.yml -i inventory.ini`	Run a playbook with a specific inventory file.
`ansible-playbook site.yml --limit webservers`	Run a playbook against a specific host group.
`ansible-playbook site.yml --tags 'deploy'`	Run only tasks matching a specific tag.
`ansible-playbook site.yml -v`	Run a playbook with verbose output.
`ansible-playbook site.yml --syntax-check`	Validate playbook syntax without execution.
`ansible-doc -l`	List all available Ansible modules.
`ansible-doc apt`	Show documentation for a specific module.
`ansible-galaxy install role_name`	Install a role from Ansible Galaxy.
`ansible-galaxy init myrole`	Scaffold a new role directory structure.
`ansible-galaxy collection install community.general`	Install an Ansible collection.
`ansible-vault create secrets.yml`	Create an encrypted secrets file.
`ansible-vault edit secrets.yml`	Edit an encrypted vault file.
`ansible-vault encrypt vars.yml`	Encrypt an existing plaintext file.
`ansible-vault decrypt vars.yml`	Decrypt an encrypted vault file.
`ansible-inventory --list`	Display the parsed inventory in JSON.
`ansible webservers -m shell -a 'uptime'`	Run an ad-hoc shell command against a host group.
`ansible all -m setup`	Gather facts about all hosts.

Command	Description
`ansible-playbook site.yml --become`	Run tasks with privilege escalation (sudo).
`ansible-config dump`	Show the current Ansible configuration.
`ansible-playbook site.yml --start-at-task='Install packages'`	Resume a playbook from a specific task.
`ansible-pull -U <repo-url>`	Pull and run a playbook from a remote repository.

9. AWS CLI — Cloud Resource Management

Commonly used AWS CLI commands for managing EC2, S3, IAM, ECS, Lambda, and CloudFormation resources.

Command	Description
`aws configure`	Set up AWS CLI credentials and default region.
`aws sts get-caller-identity`	Show the identity of the current AWS credentials.
`aws ec2 describe-instances`	List EC2 instances and their details.
`aws ec2 start-instances --instance-ids i-xxxx`	Start a stopped EC2 instance.
`aws ec2 stop-instances --instance-ids i-xxxx`	Stop a running EC2 instance.
`aws ec2 terminate-instances --instance-ids i-xxxx`	Permanently terminate an EC2 instance.
`aws ec2 describe-security-groups`	List EC2 security groups.
`aws ec2 create-key-pair --key-name mykey`	Create a new EC2 SSH key pair.
`aws s3 ls`	List S3 buckets.
`aws s3 ls s3://mybucket`	List objects in an S3 bucket.
`aws s3 cp file.txt s3://mybucket/`	Copy a file to an S3 bucket.
`aws s3 sync ./local s3://mybucket/`	Sync a local directory with an S3 bucket.
`aws s3 rm s3://mybucket/file.txt`	Delete an object from S3.
`aws s3 mb s3://mybucket`	Create a new S3 bucket.
`aws s3api get-bucket-policy --bucket mybucket`	Retrieve the policy attached to an S3 bucket.
`aws iam list-users`	List IAM users.
`aws iam create-user --user-name newuser`	Create a new IAM user.
`aws iam attach-user-policy --user-name newuser --policy-arn <arn>`	Attach a policy to an IAM user.
`aws iam list-roles`	List IAM roles.
`aws iam create-role --role-name myrole --assume-role-policy-document file://policy.json`	Create a new IAM role.

Command	Description
aws lambda list-functions	List deployed Lambda functions.
aws lambda invoke --function-name myfunc out.json	Invoke a Lambda function synchronously.
aws lambda update-function-code --function-name myfunc --zip-file fileb://code.zip	Update Lambda function code.
aws ecs list-clusters	List ECS clusters.
aws ecs list-tasks --cluster mycluster	List tasks running in an ECS cluster.
aws ecs update-service --cluster mycluster --service myservice --force-new-deployment	Force a new ECS service deployment.
aws eks list-clusters	List EKS clusters.
aws eks update-kubeconfig --name mycluster	Configure kubectl to access an EKS cluster.
aws cloudformation deploy --template-file template.yaml --stack-name mystack	Deploy a CloudFormation stack.
aws cloudformation describe-stacks	List CloudFormation stacks and their status.
aws cloudformation delete-stack --stack-name mystack	Delete a CloudFormation stack.
aws rds describe-db-instances	List RDS database instances.
aws logs tail /aws/lambda/myfunc --follow	Stream CloudWatch logs in real time.
aws cloudwatch get-metric-statistics	Retrieve metric statistics from CloudWatch.
aws route53 list-hosted-zones	List Route 53 DNS hosted zones.
aws acm list-certificates	List ACM SSL/TLS certificates.
aws ecr get-login-password | docker login --username AWS --password-stdin <registry>	Authenticate Docker to an ECR registry.
aws ecr create-repository --repository-name myrepo	Create a new ECR repository.
aws sns publish --topic-arn <arn> --message 'hello'	Publish a message to an SNS topic.

Command	Description
`aws sqs send-message --queue-url <url> --message-body 'hello'`	Send a message to an SQS queue.
`aws secretsmanager get-secret-value --secret-id mysecret`	Retrieve a secret value from Secrets Manager.

10. Azure CLI — Cloud Resource Management

Frequently used Azure CLI commands for managing resource groups, VMs, AKS, storage, and security.

Command	Description
`az login`	Authenticate the Azure CLI with your account.
`az account show`	Display details of the active subscription.
`az account set --subscription <id>`	Switch the active Azure subscription.
`az group create --name mygroup --location eastus`	Create a new resource group.
`az group delete --name mygroup`	Delete a resource group and its resources.
`az vm create --resource-group mygroup --name myvm --image UbuntuLTS`	Create a new virtual machine.
`az vm start --name myvm --resource-group mygroup`	Start a stopped virtual machine.
`az vm stop --name myvm --resource-group mygroup`	Stop a running virtual machine.
`az vm list`	List virtual machines in the subscription.
`az vm show --name myvm --resource-group mygroup`	Show detailed info about a VM.
`az aks create --resource-group mygroup --name mycluster --node-count 3`	Create a new AKS cluster.
`az aks get-credentials --resource-group mygroup --name mycluster`	Configure kubectl for an AKS cluster.
`az aks scale --name mycluster --resource-group mygroup --node-count 5`	Scale the node count of an AKS cluster.
`az aks upgrade --name mycluster --resource-group mygroup`	Upgrade an AKS cluster's Kubernetes version.
`az storage account create --name mystorage --resource-group mygroup`	Create a new storage account.
`az storage blob upload --container-name mycontainer --file file.txt`	Upload a file to blob storage.

Command	Description
`az storage blob list --container-name mycontainer`	List blobs in a storage container.
`az acr create --resource-group mygroup --name myregistry --sku Basic`	Create a new Azure Container Registry.
`az acr login --name myregistry`	Authenticate Docker to an ACR registry.
`az acr build --registry myregistry --image myapp:v1 .`	Build and push an image using ACR Tasks.
`az webapp create --resource-group mygroup --plan myplan --name myapp`	Create a new App Service web app.
`az webapp deploy --resource-group mygroup --name myapp --src-path app.zip`	Deploy code to an App Service.
`az functionapp create --resource-group mygroup --name myfunc --storage-account mystorage`	Create a new Function App.
`az network vnet create --resource-group mygroup --name myvnet`	Create a virtual network.
`az network nsg create --resource-group mygroup --name mynsg`	Create a network security group.
`az keyvault create --name mykv --resource-group mygroup`	Create a new Key Vault.
`az keyvault secret set --vault-name mykv --name mysecret --value 'value'`	Store a secret in Key Vault.
`az ad user list`	List Azure AD users.
`az role assignment create --assignee <id> --role Contributor`	Assign an RBAC role to a principal.
`az monitor metrics list --resource <id>`	Retrieve metrics for a resource.
`az deployment group create --resource-group mygroup --template-file template.json`	Deploy an ARM template to a resource group.

11. Jenkins & CI/CD Pipeline Operations

Commands and CLI operations used for managing Jenkins jobs, pipelines, and continuous integration workflows.

Command	Description
`jenkins-cli build <job-name>`	Trigger a build of a specific Jenkins job.
`jenkins-cli list-jobs`	List all jobs configured on the Jenkins server.
`jenkins-cli get-job <job-name>`	Retrieve the XML configuration of a job.
`jenkins-cli create-job <job-name> < config.xml`	Create a new job from an XML config.
`jenkins-cli delete-job <job-name>`	Delete an existing Jenkins job.
`jenkins-cli console <job-name>`	Show the console output of the last build.
`jenkins-cli safe-restart`	Restart Jenkins safely after running builds complete.
`jenkins-cli install-plugin <plugin-name>`	Install a plugin on the Jenkins server.
`jenkins-cli reload-configuration`	Reload Jenkins configuration from disk.
`curl -X POST JENKINS_URL/job/myjob/build`	Trigger a Jenkins job build via the REST API.
`systemctl status jenkins`	Check the status of the Jenkins service.
`cat /var/log/jenkins/jenkins.log`	View Jenkins server logs.
`mvn clean install`	Build a Maven project, cleaning previous artifacts first.
`mvn test`	Run unit tests for a Maven project.
`mvn package`	Package a Maven project into a JAR/WAR.
`mvn deploy`	Deploy build artifacts to a remote repository.
`gradle build`	Build a Gradle project.
`gradle test`	Run tests for a Gradle project.
`npm install`	Install Node.js project dependencies.
`npm run build`	Run the build script defined in package.json.

12. Networking & Connectivity

Networking diagnostic and configuration commands used to troubleshoot connectivity in DevOps environments.

Command	Description
`ping google.com`	Test network connectivity and latency to a host.
`traceroute google.com`	Trace the network path packets take to a destination.
`curl -I https://example.com`	Fetch only HTTP response headers from a URL.
`curl -X POST -d '{}' https://api.example.com`	Send an HTTP POST request with a JSON body.
`wget https://example.com/file.zip`	Download a file from a URL.
`nslookup example.com`	Query DNS to resolve a domain name.
`dig example.com`	Perform detailed DNS lookups.
`host example.com`	Resolve a hostname to its IP address.
`telnet host port`	Test TCP connectivity to a host and port.
`nc -zv host port`	Check if a specific port is open (netcat).
`ifconfig`	Display network interface configuration (legacy).
`ip a`	Display network interfaces and IP addresses (modern).
`ip route`	Display the system's routing table.
`route -n`	Show the kernel routing table.
`arp -a`	Display the ARP cache table.
`iptables -L`	List current firewall rules.
`ufw allow 22/tcp`	Allow traffic on a specific port through the firewall.
`ufw status`	Show the status of the uncomplicated firewall.
`firewall-cmd --list-all`	List all active firewalld rules.
`ssh user@host`	Connect to a remote host via SSH.
`ssh -i key.pem user@host`	Connect via SSH using a specific private key.
`scp file.txt user@host:/path`	Securely copy a file to a remote host.
`rsync -avz ./local user@host:/remote`	Efficiently sync files to a remote host.
`ssh-keygen -t rsa -b 4096`	Generate a new SSH key pair.
`ssh-copy-id user@host`	Copy a public SSH key to a remote host's authorized_keys.

13. Monitoring, Logging & Observability

Commands related to Prometheus, Grafana, and log management used to keep systems observable and reliable.

Command	Description
`promtool check config prometheus.yml`	Validate a Prometheus configuration file.
`promtool query instant <url> <query>`	Run an instant PromQL query against a Prometheus server.
`curl localhost:9090/-/healthy`	Check the health endpoint of a Prometheus server.
`curl localhost:9090/api/v1/query?query=up`	Query Prometheus for the 'up' metric via API.
`systemctl status prometheus`	Check the status of the Prometheus service.
`systemctl status grafana-server`	Check the status of the Grafana server.
`grafana-cli plugins install <plugin>`	Install a Grafana plugin.
`amtool alert query`	Query active alerts in Alertmanager.
`amtool config show`	Show the current Alertmanager configuration.
`node_exporter --version`	Show the installed version of node_exporter.
`journalctl -u myservice --since '1 hour ago'`	Show systemd logs for a service from a relative time.
`tail -f /var/log/nginx/access.log`	Follow Nginx access logs in real time.
`logrotate -f /etc/logrotate.conf`	Force log rotation according to configuration.
`curl -s localhost:9100/metrics`	Fetch raw metrics exposed by node_exporter.
`kubectl top pod --containers`	Show per-container resource usage within pods.

14. Package Management & Build Tools

Cross-distribution package management and language-specific build tool commands.

Command	Description
`apt update && apt upgrade -y`	Update package index and upgrade installed packages (Debian/Ubuntu).
`apt install -y package`	Install a package on Debian/Ubuntu systems.
`apt remove package`	Remove a package on Debian/Ubuntu systems.
`yum install package`	Install a package on RHEL/CentOS systems.
`yum update`	Update all packages on RHEL/CentOS systems.
`dnf install package`	Install a package using DNF (Fedora/RHEL 8+).
`snap install package`	Install a package via Snap (sandboxed packages).
`brew install package`	Install a package via Homebrew (macOS/Linux).
`pip install package`	Install a Python package via pip.
`pip freeze > requirements.txt`	Export installed Python packages to a requirements file.
`pip install -r requirements.txt`	Install Python packages from a requirements file.
`python -m venv venv`	Create a Python virtual environment.
`source venv/bin/activate`	Activate a Python virtual environment.
`npm install -g package`	Install an npm package globally.
`npm list --depth=0`	List top-level installed npm packages.
`yarn add package`	Add a package using Yarn.

15. Security, Secrets & Encryption

Commands used for handling certificates, encryption, and credential security in DevOps workflows.

Command	Description
`openssl genrsa -out key.pem 2048`	Generate a new RSA private key.
`openssl req -new -key key.pem -out request.csr`	Create a certificate signing request.
`openssl x509 -in cert.pem -text -noout`	Display details of an SSL certificate.
`openssl s_client -connect host:443`	Test and inspect an SSL/TLS connection.
`gpg --gen-key`	Generate a new GPG key pair.
`gpg --encrypt --recipient user file.txt`	Encrypt a file using a recipient's public key.
`gpg --decrypt file.txt.gpg`	Decrypt a GPG-encrypted file.
`vault login`	Authenticate to a HashiCorp Vault server.
`vault kv put secret/myapp key=value`	Store a secret in Vault's key-value store.
`vault kv get secret/myapp`	Retrieve a secret from Vault.
`sops -e secrets.yaml > secrets.enc.yaml`	Encrypt a file using SOPS.
`sops -d secrets.enc.yaml`	Decrypt a SOPS-encrypted file.
`trivy image myapp:latest`	Scan a container image for known vulnerabilities.
`kubectl get secrets -o json`	Dump all Kubernetes secrets in JSON format.

16. Miscellaneous DevOps Utilities

Additional everyday utility commands that round out a DevOps engineer's toolkit.

Command	Description
`jq '.' file.json`	Pretty-print and query JSON data on the command line.
`yq '.spec' file.yaml`	Pretty-print and query YAML data on the command line.
`base64 file.txt`	Encode a file's contents to base64.
`base64 -d encoded.txt`	Decode base64-encoded content.
`md5sum file.txt`	Generate an MD5 checksum of a file.
`sha256sum file.txt`	Generate a SHA-256 checksum of a file.
`date '+%Y-%m-%d %H:%M:%S'`	Print the current date and time in a custom format.
`cal`	Display a calendar in the terminal.
`man command`	Show the manual page for a command.
`which command`	Show the full path of a command's executable.
`type command`	Show how the shell interprets a command name.
`echo $PATH`	Print the current PATH environment variable.
`source .bashrc`	Reload shell configuration in the current session.
`diff -u file1 file2`	Show differences between two files in unified format.
`envsubst < template.yaml > output.yaml`	Substitute environment variables into a template file.